

SUWA, Japan

WMO: 476200

Lat: 36.0461N

Lon: 138.1097E

Elev: 763

StdP: 92.49

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.5	-13.8	1.2	-4.4	-12.5	1.4	-3.9	10.3	0.1	9.4	0.2	1.7	160	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	31.5	22.6	30.2	22.2	28.8	21.7	23.6	29.1	23.1	28.1	22.5	27.1	3.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.4	26.2	21.6	17.8	25.6	21.1	17.3	25.1	75.2	29.4	72.7	28.3	70.5	27.2	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 9.5	8.5	7.5	DB	-11.7	33.1	1.5	1.0	-12.8	33.8	-13.7	34.4	-14.6	35.0	-15.7	35.7	(4)
(5)			WB	-12.3	24.9	1.4	0.7	-13.4	25.4	-14.2	25.8	-15.0	26.2	-16.0	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.1	-0.1	4.2	10.2	15.8	19.7	23.9	24.8	20.3	14.0	7.6	2.1
	DBStd	9.37	2.62	3.19	3.34	3.67	2.76	2.37	2.31	2.06	3.14	3.09	3.04	2.91
	HDD10.0	1187	344	283	183	41	1	0	0	0	4	85	245	
	HDD18.3	2882	603	517	439	244	86	13	0	0	17	138	323	503
	CDD10.0	1861	0	0	2	48	181	291	431	459	310	128	12	0
	CDD18.3	514	0	0	0	0	8	54	172	201	76	4	0	0
	CDH23.3	3071	0	0	0	5	71	231	1084	1353	318	7	0	0
	CDH26.7	863	0	0	0	0	6	25	334	442	56	0	0	0
(14) Wind	WSAvg	3.0	3.0	3.2	3.7	3.6	3.3	2.8	2.6	2.7	2.9	2.9	2.9	2.9
(15) Precipitation	PrecAvg	1158	42	49	80	84	104	135	154	119	167	118	69	38
	PrecMax	1501	111	98	145	195	243	280	429	227	344	370	173	80
	PrecMin	743	2	14	29	22	46	55	19	38	38	5	7	1
	PrecStd	210	28	26	31	41	48	48	88	56	74	74	42	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	12.6	17.6	23.7	27.5	29.1	32.8	33.1	30.3	24.3	18.1	12.8
		MCWB	4.6	7.6	8.8	13.5	16.0	19.3	22.9	22.9	22.0	18.4	12.5	8.6
	2%	DB	6.8	9.4	14.9	21.1	25.4	27.2	31.5	31.7	28.3	22.2	16.0	10.3
		MCWB	3.1	4.9	7.7	12.2	15.4	18.8	22.8	22.6	21.1	16.8	11.5	7.0
	5%	DB	5.0	7.1	12.5	19.0	23.5	25.9	30.1	30.6	26.7	20.6	14.4	8.8
		MCWB	1.7	3.1	6.8	11.1	14.7	18.3	22.3	22.4	20.6	15.9	10.7	5.7
	10%	DB	3.5	5.3	10.6	17.0	21.8	24.6	28.6	29.3	25.1	19.0	12.9	7.2
		MCWB	0.6	2.1	6.1	10.7	14.2	18.2	21.8	22.2	19.9	15.1	10.0	4.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	8.2	11.0	16.7	19.2	22.1	24.6	24.5	23.1	19.5	14.6	10.2
		MCDB	7.4	11.0	13.4	18.7	22.7	26.0	30.7	30.5	28.0	22.1	16.2	11.3
	2%	WB	3.6	5.6	9.3	14.4	17.7	21.1	23.6	23.7	22.1	18.2	12.9	7.9
		MCDB	5.9	8.3	12.7	18.3	21.3	24.2	29.2	29.3	26.3	21.1	14.6	9.9
	5%	WB	2.1	3.9	8.0	12.9	16.4	20.3	23.0	23.2	21.4	16.9	11.6	6.0
		MCDB	4.4	6.1	11.2	17.0	21.1	23.5	28.1	28.4	25.1	19.7	13.5	8.1
	10%	WB	1.0	2.5	6.5	11.6	15.4	19.4	22.4	22.7	20.8	15.7	10.3	4.6
		MCDB	3.1	4.9	10.1	15.9	20.0	22.9	27.1	27.4	24.1	18.3	12.5	6.8
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	10.0	10.9	10.4	8.6	7.9	8.4	7.8	8.5	9.0	8.3
		MCDBR	10.3	12.0	13.6	14.0	13.6	10.8	10.2	10.2	9.2	9.8	10.9	10.3
	5% WB	MCWBR	7.0	7.8	7.9	6.9	6.0	4.4	3.7	3.6	3.8	5.2	6.8	7.2
		MCWBR	9.1	9.9	11.4	11.0	10.7	7.9	9.2	9.1	7.9	7.9	8.8	8.9
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.388	0.456	0.503	0.523	0.559	0.542	0.549	0.474	0.423	0.369	0.342
		taud	2.166	2.054	1.933	1.887	1.895	1.871	1.985	1.958	2.119	2.180	2.269	2.263
	Ebn at Noon	Ebn at Noon	825	829	804	789	779	749	759	743	781	786	797	808
		Edh at Noon	113	143	177	195	197	201	178	180	145	125	102	97
(44) All-Sky Solar Radiation	RadAvg	2.37	3.10	3.97	4.74	5.25	4.72	4.70	4.75	3.84	3.19	2.48	2.09	
	RadStd	0.15	0.20	0.27	0.42	0.54	0.35	0.67	0.45	0.33	0.31	0.20	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page